

Simple Laravel Project Documentation

This documentation outlines a simple Laravel project with nine models: `Category`, `Product`, `User`, `Color`, `Size`, `Cart`, `CartItem`, `Order`, and `OrderItem`. It covers project setup, model relationships, database structure, and basic CRUD functionality.

Project Overview

The project is a basic e-commerce system where:

- **Users** can browse products, add them to a cart, and place orders.
- **Products** belong to a category and can have multiple colors and sizes.
- **Categories** group products.
- **Colors** and **Sizes** allow product variations.
- **Carts** store products for users before checkout.
- **CartItem**s represent individual items in a cart with specific color and size.
- **Orders** represent completed purchases.
- **OrderItem**s detail the products in an order with specific color and size.

The project uses Laravel's Eloquent ORM for database interactions and provides basic CRUD operations for each model.

Prerequisites

- PHP >= 8.1
- Composer
- Laravel >= 10.x
- MySQL or any supported database
- Basic knowledge of Laravel and PHP

Database Structure

Tables

The project uses nine tables corresponding to the models:

- `users` : Stores user information.
- `categories` : Stores category details.
- `products` : Stores product details, linked to a category.
- `colors` : Stores available colors for products.
- `sizes` : Stores available sizes for products.
- `carts` : Stores user shopping carts.
- `cart_items` : Stores individual items in a cart, linked to product, color, and size.
- `orders` : Stores user orders.
- `order_items` : Stores individual items in an order, linked to product, color, and size.

Schema

- **users** (default Laravel table):
 - `id` : Primary key
 - `name` : User's name (string)
 - `email` : User's email (string, unique)
 - `password` : Hashed password
 - `created_at`, `updated_at` : Timestamps
- **categories**:
 - `id` : Primary key
 - `name` : Category name (string, unique)
 - `description` : Category description (text, nullable)
 - `created_at`, `updated_at` : Timestamps
- **products**:
 - `id` : Primary key
 - `name` : Product name (string)
 - `description` : Product description (text, nullable)
 - `price` : Product price (decimal)
 - `category_id` : Foreign key referencing `categories(id)`
 - `created_at`, `updated_at` : Timestamps
- **colors**:
 - `id` : Primary key
 - `name` : Color name (string, unique)
 - `created_at`, `updated_at` : Timestamps
- **sizes**:

- `id` : Primary key
 - `name` : Size name (string, unique, e.g., "S", "M", "L")
 - `created_at`, `updated_at` : Timestamps
- **carts:**
 - `id` : Primary key
 - `user_id` : Foreign key referencing `users(id)`
 - `created_at`, `updated_at` : Timestamps
- **cart_items:**
 - `id` : Primary key
 - `cart_id` : Foreign key referencing `carts(id)`
 - `product_id` : Foreign key referencing `products(id)`
 - `color_id` : Foreign key referencing `colors(id)`
 - `size_id` : Foreign key referencing `sizes(id)`
 - `quantity` : Quantity of the product (integer)
 - `created_at`, `updated_at` : Timestamps
- **orders:**
 - `id` : Primary key
 - `user_id` : Foreign key referencing `users(id)`
 - `total` : Total order amount (decimal)
 - `status` : Order status (string, e.g., "pending", "completed")
 - `created_at`, `updated_at` : Timestamps
- **order_items:**
 - `id` : Primary key
 - `order_id` : Foreign key referencing `orders(id)`
 - `product_id` : Foreign key referencing `products(id)`
 - `color_id` : Foreign key referencing `colors(id)`
 - `size_id` : Foreign key referencing `sizes(id)`
 - `quantity` : Quantity of the product (integer)
 - `price` : Price at the time of purchase (decimal)
 - `created_at`, `updated_at` : Timestamps

Models and Relationships

User Model

- **File:** `app/Models/User.php`
- **Description:** Laravel's default User model.
- **Relationships:**
 - Has many `Cart` (one-to-many)
 - Has many `Order` (one-to-many)

Category Model

- **File:** `app/Models/Category.php`
- **Description:** Represents a product category.
- **Relationships:**
 - Has many `Product` (one-to-many)

Product Model

- **File:** `app/Models/Product.php`
- **Description:** Represents a product.
- **Relationships:**
 - Belongs to `Category` (many-to-one)
 - Belongs to many `Color` (many-to-many)
 - Belongs to many `Size` (many-to-many)
 - Has many `CartItem` (one-to-many)
 - Has many `OrderItem` (one-to-many)

Color Model

- **File:** `app/Models/Color.php`
- **Description:** Represents a color option for products.
- **Relationships:**
 - Belongs to many `Product` (many-to-many)
 - Has many `CartItem` (one-to-many)
 - Has many `OrderItem` (one-to-many)

Size Model

- **File:** `app/Models/Size.php`
- **Description:** Represents a size option for products.
- **Relationships:**
 - Belongs to many `Product` (many-to-many)
 - Has many `CartItem` (one-to-many)
 - Has many `OrderItem` (one-to-many)

Cart Model

- **File:** `app/Models/Cart.php`

Description: Represents a user's shopping cart.

- **Relationships:**
 - Belongs to `User` (many-to-one)
 - Has many `CartItem` (one-to-many)

CartItem Model

- **File:** `app/Models/CartItem.php`
- **Description:** Represents an item in a cart with specific product, color, and size.
- **Relationships:**
 - Belongs to `Cart` (many-to-one)
 - Belongs to `Product` (many-to-one)
 - Belongs to `Color` (many-to-one)
 - Belongs to `Size` (many-to-one)

Order Model

- **File:** `app/Models/Order.php`
- **Description:** Represents a completed order.
- **Relationships:**
 - Belongs to `User` (many-to-one)
 - Has many `OrderItem` (one-to-many)

OrderItem Model

- **File:** `app/Models/OrderItem.php`
- **Description:** Represents an item in an order with specific product, color, and size.
- **Relationships:**
 - Belongs to `Order` (many-to-one)
 - Belongs to `Product` (many-to-one)
 - Belongs to `Color` (many-to-one)
 - Belongs to `Size` (many-to-one)

Notes for Developers

- **Validation:** Implement validation in controllers for all models (e.g., unique names for `Color` and `Size`, positive quantities for `CartItem` and `OrderItem`).
- **Filtering:** Add search and filter functionality for products (e.g., by category, color, or size).
- **Checkout Process:** Implement logic to convert `CartItem` entries to `OrderItem` entries upon order placement.
- **Testing:** Test all APIs on Postman.